AI Act Compliance

PlantHealthMonitor

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1. Version Information

Project Name : PlantHealthMonitor

Version : Prototype

Document Version : 1.0

2. Purpose of this Document

This document assesses the PlantHealthMonitor system against the requirements of Regulation (EU) 2024/1689 (Artificial Intelligence Act).

The purpose of this assessment is to:

- Determine whether PlantHealthMonitor qualifies as an AI system under the AI Act.
- Identify which articles and annexes of the AI Act are applicable.
- Assess whether the system is prohibited or high-risk.
- Define the resulting risk classification and its obligations.

The AI Act is a risk-based regulation. Therefore, only provisions relevant to the system's intended purpose and risk level are assessed in detail.

3. System Description

PlantHealthMonitor is an embedded IoT system designed to help users monitor and care for their indoor plants.

The system collects environmental sensor data from an ESP32 microcontroller connected to a set of sensors. Based on the collected data, the system classifies the plant's current health condition and provides care recommendations to the user via a mobile and desktop application.

The system is designed as a support tool that assists users in understanding their plant's condition and does not make decisions on behalf of users.

4. Intended Purpose

The primary purpose of PlantHealthMonitor is to:

- Monitor environmental conditions around an indoor plant.
- Classify the plant's health as healthy, moderate risk, or high risk.
- Predict the time until a care action (e.g. watering) becomes urgent.
- Provide actionable care recommendations to the user.
- Notify the user when abnormal plant conditions are detected.

The system is intended exclusively for personal indoor plant care.

The system is not intended for:

- Employee monitoring.
- Educational assessment.
- Healthcare diagnosis.
- Creditworthiness assessment.
- Legal decision-making.
- Government or law enforcement purposes.

5. System Functionality

5.1. Input Data

The system processes the following categories of environmental data:

- Soil moisture percentage.
- Ambient temperature in degrees Celsius.
- Relative humidity percentage.
- Light intensity percentage.
- Historical sensor readings from the preceding measurement cycles.

No biometric data, personal data, or user-identifying information is collected or processed.

5.2. System Outputs

The system generates outputs including:

- A health risk classification (healthy / moderate risk / high risk).
- A time estimate until a care action is needed.
- Boolean care action flags (water, reduce temperature, increase light, increase humidity).
- A human-readable recommendation summary.
- Push notifications when risk class changes to moderate or high.

5.3. AI Functionality

5.3.1. Current Functionality

The current functionality includes:

- On-device health risk classification using a trained MLP neural network (TensorFlowLite).
- Predictive irrigation modelling using a trained MLP regression model.
- Rule-based care action determination informed by ML risk output.
- Per-plant preference profile support (low / mid / high per sensor).

5.3.2. Future Functionality

Future versions of the system may be able to:

- Retrain models on real collected sensor data as it accumulates.
- Expand the plant species library with pre-configured care profiles.

All recommendations provided by the system are always intended as optional guidance.

6. AI Act Assessment

The AI Act establishes a risk-based regulatory framework, meaning that different legal obligations apply depending on the nature and risk level of an AI system.

To support this assessment, the EU AI Act Compliance Checker was used as an indicative tool to identify applicable provisions.

PlantHealthMonitor qualifies as an AI system under Article 3(1) as it generates predictions and recommendations based on sensor data. Having been assessed against the prohibited practices of Article 5 and the high-risk domains of Annex III, the system does not fall under either category - it monitors plant conditions for personal use, processes no personal or biometric data, and produces no legally significant decisions.

The assessment therefore focuses on the articles that are directly applicable to the system.

6.1. Risk Classification

PlantHealthMonitor is classified as a **Minimal-Risk AI System** under the AI Act. The system provides health classifications and care recommendations but does not operate in any high-risk domain, does not engage in prohibited AI practices, and does not trigger Article 50 limited-risk transparency obligations.

6.2. Article 69 - Codes of Conduct for Minimal-Risk Systems

Article 69 encourages providers of minimal-risk AI systems to voluntarily adopt codes of conduct and apply best practices in transparency, robustness, and human oversight. PlantHealthMonitor applies these principles voluntarily as described in the sections below.

7. Transparency Measures

In the spirit of Article 50 and Article 13, and in line with Article 69 (voluntary codes of conduct for minimal-risk systems), the system ensures transparency through:

- Displaying raw sensor readings alongside every recommendation so users understand the basis for each suggestion.
- Explicitly labelling the risk classification (Healthy / Moderate Risk / High Risk) rather than hiding it inside an opaque output.
- Communicating care actions as suggestions, not instructions.
- Providing documentation (this document, the user manual, and the final project report) explaining the model architecture, training data, and limitations in plain language.

8. Human Oversight

Human oversight is maintained at all times.

The system:

- Does not autonomously make decisions.
- Does not enforce care actions.
- Does not physically actuate any hardware (no automated watering in the current implementation).
- Does not modify plant settings automatically.

Users retain full control over all decisions and may ignore or disable recommendations at any time.

9. Risk Assessment

Risk	Likelihood	Impact	Mitigation
Incorrect health risk classification	Medium	Low	Outputs are advisory; user can observe plant directly.
Inaccurate time-to-watering prediction	Medium	Low	Presented as an estimate; user retains final decision.
Sensor hardware failure producing misleading readings	Low	Low	User advised to periodically verify readings appear reasonable.
Overreliance on recommendations	Low	Medium	No enforcement mechanisms; system frames outputs as suggestions.
Model inaccuracy for uncommon plant species	Medium	Low	Synthetic training data limitation documented; user can adjust preferences.

10. Technical Limitations

The system has several known limitations:

- Models are trained on synthetic data; predictions may be less accurate for conditions outside the training distribution.
- Model accuracy improves as real-world data accumulates and retraining occurs.
- The 3-hour measurement interval means short-lived stress events between readings may not be detected.
- Sensor hardware failures cannot be detected by the system.
- Recommendation quality may vary across plant species with different care profiles.

Outputs should be interpreted as informational guidance rather than factual determinations.

11. Conclusion

PlantHealthMonitor is an AI-enabled plant monitoring system that analyses environmental sensor data and generates health classifications and care recommendations.

The system qualifies as an AI system under Article 3(1) of Regulation (EU) 2024/1689.

It does not involve prohibited AI practices under Article 5 and does not fall within any high-risk category defined in Annex III.

The system is therefore classified as a **Minimal-Risk AI System**, with no mandatory compliance obligations under the Act. Transparency measures and human oversight provisions have been voluntarily implemented in line with Article 69 and the broader intent of the regulation.

12. References

- EU AI Act Compliance Checker EU Artificial Intelligence Act. (z.d.). https://artificialintelligenceact.eu/assessment/eu-ai-act-compliance-checker/
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